Problem 1

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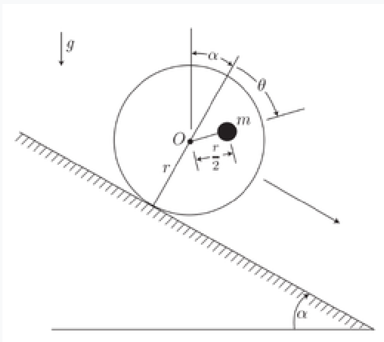
A disk of radius r and negligible mass (i.e., assume the disk is massless) has a particle of mass m embedded at a distance $\frac{1}{2}r$ from the center O, as shown in the figure. Assume the disk rolls without slipping down a plane inclined at an angle α from the horizontal. Use θ as the single degree of freedom for this system. Derive the Hamilton's equations of motion for θ, p_θ.



Setup

Let $\hat{n}_1$ point down the incline (rolling direction) and $\hat{n}_2$ be the outward normal to the incline surface. Rolling without slipping ties the center O's translation to θ:

$$\vec{r}_O = r\theta\,\hat{n}_1 + r\,\hat{n}_2$$

The embedded mass, rotating rigidly with the disk at radius $r/2$, is at

$$\vec{r}_m = \vec{r}_O + \frac{r}{2}\sin\theta\,\hat{n}_1 + \frac{r}{2}\cos\theta\,\hat{n}_2 = \left(r\theta + \frac{r}{2}\sin\theta\right)\hat{n}_1 + \left(r + \frac{r}{2}\cos\theta\right)\hat{n}_2$$

Kinetic and Potential Energy

$$\dot{\vec{r}}_m = \left(r\dot{\theta} + \frac{r}{2}\dot{\theta}\cos\theta\right)\hat{n}_1 - \frac{r}{2}\dot{\theta}\sin\theta\,\hat{n}_2$$

$$|\dot{\vec{r}}_m|^2 = r^2\dot{\theta}^2\left(1 + \frac{1}{2}\cos\theta\right)^2 + \frac{r^2}{4}\dot{\theta}^2\sin^2\theta = r^2\dot{\theta}^2\left(\frac{5}{4} + \cos\theta\right)$$

$$\boxed{T = \frac{1}{2}I(\theta)\dot{\theta}^2, \qquad I(\theta) \equiv mr^2\left(\frac{5}{4} + \cos\theta\right)}$$

With $\vec{g} = g\sin\alpha\,\hat{n}_1 - g\cos\alpha\,\hat{n}_2$ and $V = -m\vec{g}\cdot\vec{r}_m$:

$$\boxed{V(\theta) = mgr\left[\cos\alpha\left(1 + \frac{1}{2}\cos\theta\right) - \sin\alpha\left(\theta + \frac{1}{2}\sin\theta\right)\right]}$$

Lagrangian, Momentum, and Hamiltonian

$$L = T - V = \frac{1}{2}I(\theta)\dot{\theta}^2 - V(\theta), \qquad p_\theta = \frac{\partial L}{\partial\dot{\theta}} = I(\theta)\dot{\theta}$$

L is time-independent and T is homogeneous quadratic in $\dot\theta$, so $H = T + V$, with $\dot\theta = p_\theta / I(\theta)$:

$$H(\theta, p_\theta) = \frac{p_\theta^2}{2mr^2\left(\frac{5}{4} + \cos\theta\right)} + mgr\left[\cos\alpha\left(1 + \frac{1}{2}\cos\theta\right) - \sin\alpha\left(\theta + \frac{1}{2}\sin\theta\right)\right]$$

Hamilton's Equations of Motion

$$\dot\theta = \frac{\partial H}{\partial p_\theta}, \qquad \dot p_\theta = -\frac{\partial H}{\partial \theta}$$

$$\dot\theta = \frac{p_\theta}{mr^2\left(\frac{5}{4} + \cos\theta\right)}$$

$$\dot p_\theta = -\frac{p_\theta^2 \sin\theta}{2mr^2\left(\frac{5}{4} + \cos\theta\right)^2} + mgr\left[\sin\alpha\left(1 + \frac{1}{2}\cos\theta\right) + \frac{1}{2}\cos\alpha\sin\theta\right]$$

Problem 2

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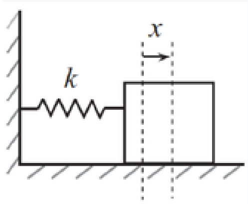

When we use Hamiltonians, we can define canonical coordinate changes (under certain conditions) from (q, p) to (Q, P) by using *generating functions* $F(q, Q, t)$. With this function in terms of the old and new coordinates, the conjugate momenta are defined by,

$$p = \frac{\partial F}{\partial q}, \quad P = -\frac{\partial F}{\partial Q}.$$

Using these relationships, we can write a new Hamiltonian $H'(Q, P)$, which follows the same canonical equations of motion.

$$\dot{Q} = \frac{\partial H'}{\partial P}, \quad \dot{P} = -\frac{\partial H'}{\partial Q}$$

As an example, consider a simple harmonic oscillator with mass $m = 1$ and a spring coefficient k. Let $\omega = \sqrt{k}$. Let $q = x$ be our only generalized coordinate.

(a) Derive the Hamiltonian for this system in terms of q, p, ω in the usual way. You should find that,

$$H(q, p) = \frac{1}{2}\left(p^2 + \omega^2 q^2\right).$$

(b) Choose the following generating function (I know it seems ridiculous, trust me):

$$F(q, Q) = \frac{1}{2}\omega q^2 cot(2\pi Q)$$

(c) Use the relationships above to derive $p(q, Q)$ and $P(q, Q)$.

(d) From the expression for P, you solve for $q(Q, P)$, and then substitute this into the expression for p to find $p(Q, P)$.

(e) Now find $H'(Q, P) = H(q(Q, P), p(Q, P))$. ($H'$ is a new Hamiltonian in terms of Q and P, not a derivative.)

(f) Write the equations of motion of Q, P. You will find that $\dot{Q}$ =constant and $\dot{P} = 0$!

(g) Integrate to find $Q(t), P(t)$ and plug into $q(Q, P), p(Q, P)$. You will have found functions of time (in terms of the constant P)! (One textbook author calls this cracking a peanut with a sledgehammer.")

There are three other types of generating functions ($F(q, P), F(p, Q), F(p, P)$) that can be used to canonically change coordinates under certain conditions. We won't fully go into these approaches in this class, but I wanted to show a simple example.

Part (a): Hamiltonian

With $q = x$, $m = 1$, $\omega = \sqrt{k}$:

$$T = \frac{1}{2}\dot{x}^2, \qquad V = \frac{1}{2}kx^2 = \frac{1}{2}\omega^2 x^2, \qquad p_x = \frac{\partial L}{\partial \dot{x}} = \dot{x}$$

$$\boxed{H(x, p_x) = T + V = \frac{1}{2}\left(p_x^2 + \omega^2 x^2\right)}$$

Part (b): Generating Function

$$F(q, Q) = \frac{1}{2}\omega q^2 \cot(2\pi Q)$$

Part (c): $p(q,Q)$ and $P(q,Q)$

$$p = \frac{\partial F}{\partial q} = \omega q \cot(2\pi Q), \qquad P = -\frac{\partial F}{\partial Q} = -\frac{1}{2}\omega q^2\big(-2\pi \csc^2(2\pi Q)\big)$$

$$\boxed{p(q,Q) = \omega q \cot(2\pi Q), \qquad P(q,Q) = \frac{\pi \omega q^2}{\sin^2(2\pi Q)}}$$

Part (d): $p(Q,P)$

Solving $P(q,Q)$ for q:

$$\boxed{q(Q,P) = \sqrt{\frac{P}{\pi\omega}}\sin(2\pi Q)}$$

Substituting into $p = \omega q \cot(2\pi Q)$:

$$p = \omega \sqrt{\frac{P}{\pi\omega}}\sin(2\pi Q) \cdot \frac{\cos(2\pi Q)}{\sin(2\pi Q)}$$

$$\boxed{p(Q,P) = \sqrt{\frac{\omega P}{\pi}}\cos(2\pi Q)}$$

Part (e): $H'(Q,P)$

$$p^2 = \frac{\omega P}{\pi}\cos^2(2\pi Q), \qquad \omega^2 q^2 = \frac{\omega P}{\pi}\sin^2(2\pi Q) \qquad \Longrightarrow \qquad p^2 + \omega^2 q^2 = \frac{\omega P}{\pi}$$

$$\boxed{H'(Q,P) = \frac{1}{2}(p^2 + \omega^2 q^2) = \frac{\omega P}{2\pi}}$$

Part (f): Equations of Motion of Q, P

$$\boxed{\dot{Q} = \frac{\partial H'}{\partial P} = \frac{\omega}{2\pi} = \text{const}, \qquad \dot{P} = -\frac{\partial H'}{\partial Q} = 0}$$

Q is cyclic in H', so P is conserved.

Part (g): Integration and Recovery of $q(t), p(t)$

$$P(t) = P_0, \qquad Q(t) = \frac{\omega}{2\pi}t + Q_0$$

Substituting into $q(Q,P)$, $p(Q,P)$ (so that $2\pi Q(t) = \omega t + 2\pi Q_0$):

$$\boxed{q(t) = \sqrt{\frac{P_0}{\pi\omega}} \sin\left(\omega t + 2\pi Q_0\right), \qquad p(t) = \sqrt{\frac{\omega P_0}{\pi}} \cos\left(\omega t + 2\pi Q_0\right)}$$

which is exactly the SHO solution, with amplitude $\sqrt{P_0/(\pi\omega)}$ and phase $2\pi Q_0$ set by the conserved P_0 and constant of integration Q_0.

Problem 3

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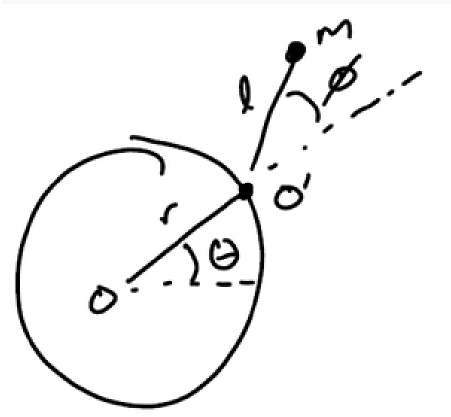
A particle of mass m is attached by a string of length l to a point O' which moves around a hoop of radius r at a constant angular rate $\dot{\theta} = \omega_0$. The angle between the radial direction and the string is given by ϕ. This setup is moving horizontally on a table top (i.e., gravity points into the page).

Apply Hamilton's Equations of Motion for the generalized coordinate $q = \phi$ and its conjugate momentum p.



Setup

Let θ (with $\dot{\theta} = \omega_0$) locate O' on the hoop and ϕ the angle of the string from the radial direction OO', so the string's absolute orientation is $\theta + \phi$:

$$x = r\cos\theta + l\cos(\theta+\phi), \qquad y = r\sin\theta + l\sin(\theta+\phi)$$

Since the tabletop is horizontal, gravity acts perpendicular to the plane of motion and $V = 0$.

Kinetic Energy

With $\dot{\theta} = \omega_0$ constant,

$$\dot{x} = -r\omega_0\sin\theta - l(\omega_0+\dot{\phi})\sin(\theta+\phi), \qquad \dot{y} = r\omega_0\cos\theta + l(\omega_0+\dot{\phi})\cos(\theta+\phi)$$

Using $\sin\theta\sin(\theta+\phi) + \cos\theta\cos(\theta+\phi) = \cos\phi$:

$$|\dot{\vec{r}}|^2 = \dot{x}^2 + \dot{y}^2 = r^2\omega_0^2 + l^2(\omega_0+\dot{\phi})^2 + 2rl\,\omega_0(\omega_0+\dot{\phi})\cos\phi$$

$$\boxed{T = L = \frac{1}{2}ml^2\dot{\phi}^2 + ml\,\omega_0(l + r\cos\phi)\,\dot{\phi} + \frac{m\omega_0^2}{2}\left(r^2 + l^2 + 2rl\cos\phi\right), \qquad V = 0}$$

Momentum and Hamiltonian

$$p_\phi = \frac{\partial L}{\partial\dot{\phi}} = ml^2\dot{\phi} + ml\,\omega_0(l + r\cos\phi)$$

The transformation to ϕ is rheonomic (O' moves with prescribed $\theta = \omega_0 t$), so T is not homogeneous quadratic in $\dot{\phi}$. Splitting $T = T_2 + T_1 + T_0$ by degree in $\dot{\phi}$,

$$T_2 = \frac{1}{2} ml^2 \dot{\phi}^2, \qquad T_1 = ml\,\omega_0 (l + r\cos\phi)\dot{\phi}, \qquad T_0 = \frac{m\omega_0^2}{2}\left(r^2 + l^2 + 2rl\cos\phi\right)$$

the general result $H = p_\phi \dot{\phi} - L = T_2 - T_0 + V$ (with $V = 0$) applies here, not $H = T + V$:

$$\boxed{H(\phi, \dot{\phi}) = T_2 - T_0 = \frac{1}{2} ml^2 \dot{\phi}^2 - \frac{m\omega_0^2}{2}\left(r^2 + l^2 + 2rl\cos\phi\right)}$$

L itself has no explicit t-dependence once $\theta(t) = \omega_0 t$ is substituted, so H is still conserved — it is the Jacobi integral, not the total mechanical energy. Inverting the momentum relation for $\dot{\phi} = \left[p_\phi - ml\,\omega_0 (l + r\cos\phi)\right]/(ml^2)$ and substituting:

$$\boxed{H(\phi, p_\phi) = \frac{\left[p_\phi - ml\,\omega_0 (l + r\cos\phi)\right]^2}{2ml^2} - \frac{m\omega_0^2}{2}\left(r^2 + l^2 + 2rl\cos\phi\right)}$$

Hamilton's Equations of Motion

$$\dot{\phi} = \frac{\partial H}{\partial p_\phi}, \qquad \dot{p}_\phi = -\frac{\partial H}{\partial \phi}$$

$$\boxed{\dot{\phi} = \frac{p_\phi - ml\,\omega_0 (l + r\cos\phi)}{ml^2}, \qquad \dot{p}_\phi = -mrl\,\omega_0 (\omega_0 + \dot{\phi})\sin\phi}$$

Differentiating the momentum relation in time and equating to $\dot{p}_\phi$ above, the $\dot{\phi}\sin\phi$ terms cancel identically, collapsing the pair to a single second-order equation:

$$\boxed{\ddot{\phi} = -\frac{r\omega_0^2}{l}\sin\phi}$$

a simple-pendulum equation driven by the hoop's rotation rate in place of gravity, consistent with the horizontal tabletop ($V = 0$).